S181M116 Derivatives and Alternative Investments: Lecture 1

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This lecture introduces the basic language and market architecture of derivatives. We study forwards, futures, options, exchanges, OTC markets, clearing houses, margin accounts, and the first no-arbitrage arguments. We also use major derivatives mishaps to establish a risk-management discipline that will remain important throughout the course.

> **Reading.** Hull, Chapters 1, 2, and selected lessons from Chapter 37. The goal is not to memorize institutional detail. The goal is to understand how contract design, clearing, collateral, and no-arbitrage logic determine who bears which risks.

What Is a Derivative?

A *derivative* is a financial contract whose value depends on an underlying variable. The underlying may be a traded asset price, such as a stock price, bond price, commodity price, exchange rate, or stock index. It may also be a rate, spread, credit event, weather variable, or other observable quantity.

Definition 1 (Derivative)
A derivative contract specifies cash flows at one or more future dates as a function of one or more underlying variables. If X_T denotes the relevant underlying variable at date T, the derivative payoff can often be written as

$$g(X_T),$$

where $g(\cdot)$ is the payoff function.

This definition is deliberately broad. A forward contract has payoff linear in the future asset price. A call option has a nonlinear payoff. A swap is a portfolio of future exchanges of cash flows. A credit default swap pays when a credit event occurs. The common element is that the contract derives its value from something else.

There are three reasons derivatives matter.

1. **Risk transfer.** A firm can transfer commodity, interest-rate, foreign-exchange, or equity risk to another party.
2. **Price discovery and leverage.** Derivatives allow market participants to take positions with limited initial cash outlay.
3. **Replicating logic.** Derivative prices are tightly connected to prices of underlying assets, interest rates, and other derivatives.

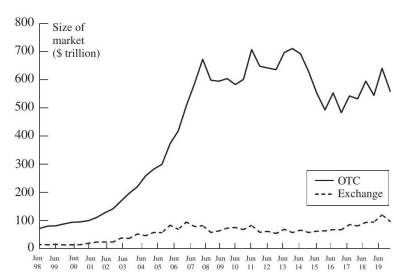

Figure 1: Derivative markets are large relative to underlying cash markets. The notional amount is not the same as market value, but it measures scale and interconnectedness.

Important distinction. A derivative can reduce risk for one party and increase risk for another. The contract itself is not "safe" or "dangerous." Its effect depends on the position, leverage, collateral, liquidity, and the institution's ability to manage the exposure.

Payoff, Profit, and Value

We will use three related but different concepts throughout the course.

Payoff The cash flow at maturity, usually written as a function of the terminal underlying price.

Profit The payoff net of the initial cost and any financing cost.

Value The market value of the contract today, before maturity.

For a contract with payoff $g(S_T)$ at maturity T, the value today is not generally $g(S_0)$. Valuation discounts, risk-adjusts, or replicates the future payoff. The central intellectual move in derivatives is to ask:

Can this future payoff be replicated by trading other assets?

If yes, the derivative's price is pinned down by no-arbitrage. If not exactly, pricing requires a model, a convention, or a market quote.

Market Architecture

Exchange-Traded Markets

An exchange-traded derivative is standardized by an exchange. The exchange specifies the underlying asset, contract size, maturity cycle, delivery or cash settlement rules, tick size, and trading procedures. Standardization makes anonymous trading and liquidity easier.

The clearing house is central. Once a trade is matched, the clearing house becomes the buyer to every seller and the seller to every buyer. This process is called *novation*. It replaces bilateral counterparty exposure with exposure to the clearing house.

Exchange-traded derivative. The trader cares less about the identity of the counterparty because the clearing house stands between the two sides. Credit risk is controlled through margin, daily settlement, default funds, and member capital.

Over-the-Counter Markets

An over-the-counter (OTC) derivative is negotiated privately. OTC contracts are more flexible than exchange-traded contracts: maturity, notional, payoff, and collateral terms can be customized. The cost of customization is weaker standardization and potentially greater counterparty, liquidity, and valuation risk.

Modern OTC markets often use central counterparties for standardized trades, but many contracts remain bilateral. Bilateral OTC trading is governed by legal agreements that define close-out netting, collateral, default events, and valuation procedures.

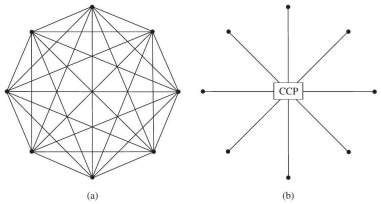

Figure 2: Bilateral OTC links create a network of counterparty exposures. A CCP changes the network but concentrates risk management in one institution.

Notional Amount Versus Market Value

The *notional principal* is the reference amount used to compute payments. It is not usually the amount at risk. For example, in a forward contract to exchange GBP 1 million for dollars, the notional is GBP 1 million. The market value may be only a small fraction of that amount because only the difference between the contracted exchange rate and the current forward rate is at stake.

Nevertheless, notional matters because it measures scale. A small percentage move on a large notional can produce a large gain or loss.

Forward Contracts

Definition and Payoffs

A forward contract is an agreement to buy or sell an asset at a specified future date T for a specified delivery price K. The party that agrees to buy has a *long forward position*. The party that agrees to sell has a *short forward position*.

Definition 2 (Forward payoff)
For one unit of the underlying asset, the payoff at maturity is

$$\textit{Long forward payoff} = S_T - K,$$

and

$$\textit{Short forward payoff} = K - S_T.$$

The payoff is linear. This linearity is what makes forwards useful for hedging. If a firm knows it must buy an asset in the future, a long forward locks in the purchase price. If a firm knows it must sell an asset in the future, a short forward locks in the sale price.

Example 1 (Foreign-exchange hedge)
A U.S. firm must pay GBP 1 million in six months. The six-month forward quote is USD 1.2230 per GBP. By entering a long forward on GBP, the firm commits to pay USD 1.223 million for GBP 1 million at maturity.

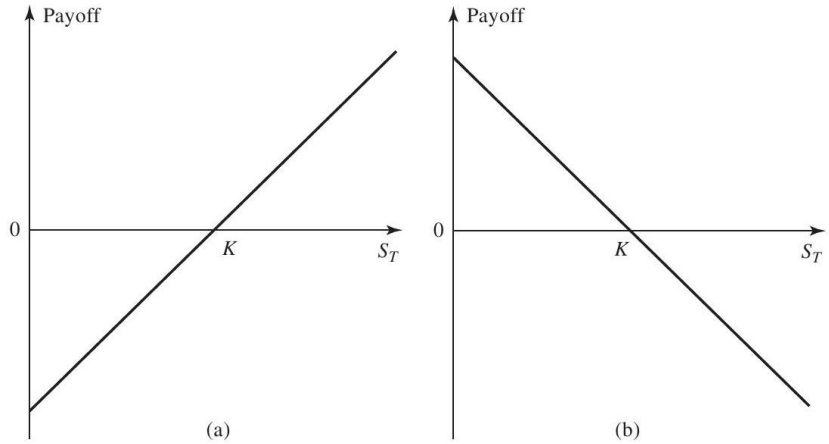

Figure 3: Forward payoffs. A long forward gains when the terminal spot price is above the delivery price. A short forward gains when it is below the delivery price.

If the future spot exchange rate is 1.3000, the forward payoff is

$$(1.3000 - 1.2230) \times 1{,}000{,}000 = \$77{,}000.$$

If the future spot exchange rate is 1.2000, the payoff is

$$(1.2000 - 1.2230) \times 1{,}000{,}000 = -\$23{,}000.$$

The hedge removes uncertainty about the dollar cost, but it also removes the benefit of a favorable exchange-rate movement.

Forward Price Versus Delivery Price

The *delivery price* K is the price written into an existing contract. The *forward price* F_0 is the delivery price that would make a newly negotiated forward contract have zero value today.

At initiation, a standard forward usually has zero value:

$$f_0 = 0.$$

That does not mean the contract is worthless economically. It means neither side pays an upfront premium because the agreed delivery price is set fairly relative to current market conditions.

A First No-Arbitrage Derivation

Consider a non-dividend-paying stock with current price S_0. Suppose the risk-free continuously compounded interest rate is r and maturity is T. We derive the no-arbitrage forward price.

Theorem 1 (Forward price for a non-income asset)
For an asset with no income and no storage cost, under frictionless markets,

$$F_0 = S_0 e^{rT}.$$

Proof 1
First suppose the quoted forward price is too high:

$$F_0 > S_0 e^{rT}.$$

Construct the following strategy at time 0:

1. *Borrow S_0 at the risk-free rate.*
2. *Buy one share of the stock.*
3. *Enter a short forward contract to sell the stock for F_0 at T.*

At time T, deliver the stock into the forward contract and receive F_0. Repay the loan, which has grown to $S_0 e^{rT}$. The terminal profit is

$$F_0 - S_0 e^{rT} > 0.$$

There was no initial net investment. This is an arbitrage.

Now suppose the quoted forward price is too low:

$$F_0 < S_0 e^{rT}.$$

If short selling is possible, construct the reverse strategy:

1. *Short sell one share and receive S_0.*
2. *Invest S_0 at the risk-free rate.*
3. *Enter a long forward contract to buy the stock for F_0 at T.*

At time T, the investment is worth $S_0 e^{rT}$. Use F_0 to buy the stock under the forward contract and return it to close the short sale. The terminal profit is

$$S_0 e^{rT} - F_0 > 0.$$

Again, this is an arbitrage. Therefore the only no-arbitrage price is $F_0 = S_0 e^{rT}$.

> **Intuition.** A forward contract postpones payment for the asset. If the asset pays no income, the forward price should equal today's spot price grown at the financing rate. Otherwise, traders can choose between "buy now and finance" and "agree to buy later" to lock in a profit.

Futures Contracts

How Futures Differ from Forwards

A futures contract is also an agreement to buy or sell an asset at a future date for a price agreed today. The key differences are institutional:

- Futures are exchange-traded and standardized.
- Futures are marked to market daily.
- Traders post initial margin and must maintain a minimum margin balance.
- Most futures positions are closed out before delivery.
- The clearing house becomes the counterparty to both sides.

Feature	Forward	Futures
Trading venue	OTC	Exchange
Terms	Customized	Standardized
Counterparty risk	Bilateral or CCP	Clearing house
Settlement	Usually at maturity	Daily mark-to-market
Initial cash flow	Usually none	Margin deposit
Closing position	Negotiated offset	Exchange offset is easy

Table 1: Forward and futures contracts have similar economic exposure but very different market plumbing.

Convergence to the Spot Price

As the delivery period approaches, the futures price must converge to the spot price of the deliverable asset, after adjusting for any contract-specific delivery options or quality differences.

The intuition is arbitrage. Suppose at delivery the futures price were above the spot price. A trader could buy the asset in the spot market and deliver it into the futures contract at the higher futures price. Suppose instead the futures price were below the spot price. A trader who needs the asset could obtain it through the futures contract more cheaply than in the spot market. Competitive trading pressure closes the gap.

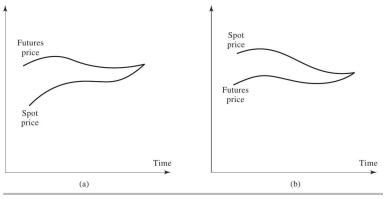

Figure 4: Convergence links the futures price to the spot price near delivery.

> **Convergence principle.** Delivery is rare in many futures markets, but the possibility of delivery is what anchors the futures price to the spot price. Without a credible settlement mechanism, a futures price could drift away from the underlying cash market.

Margin Accounts and Daily Settlement

Margin is not a down payment on the asset. It is collateral that protects the clearing system against default.

Let one futures contract be for Q units of the underlying. Let F_t denote the settlement futures price at the end of day t. For a long position in N contracts, the daily gain or loss is

$$\text{P\&L}_t = NQ(F_t - F_{t-1}).$$

For a short position, the sign is reversed:

$$\text{P\&L}_t^{\text{short}} = NQ(F_{t-1} - F_t).$$

If M_t is the margin account balance after daily settlement and before any margin call contribution, then

$$M_t = M_{t-1} + \text{P\&L}_t.$$

If M_t falls below the maintenance margin, the trader receives a margin call and must restore the account to the initial margin level.

Example 2 (Gold futures margin)
Suppose a trader is long two gold futures contracts. Each contract is for 100 ounces, so $NQ = 200$ ounces. The initial futures price is USD 1,750 per ounce. If the next settlement price is USD 1,741, the long position loses

$$200(1{,}741 - 1{,}750) = -\$1{,}800.$$

The short side receives the same amount. Futures are zero-sum before transaction costs, but the timing of cash flows differs because gains and losses are settled daily.

> **Why daily settlement matters.** A forward contract accumulates gains and losses until maturity. A futures contract realizes gains and losses every day. This reduces counterparty credit risk, but it creates liquidity risk: a trader can be economically right in the long run and still fail because margin calls arrive before convergence.

Options

Rights, Obligations, and Payoffs

A call option gives the holder the right, but not the obligation, to buy the underlying asset for strike price K. A put option gives the holder the right, but not the obligation, to sell the underlying asset for K.

Definition 3 (Option payoffs)
For a European option expiring at T,

$$\textit{Call payoff} = \max(S_T - K, 0) = (S_T - K)^+,$$

and

$$\textit{Put payoff} = \max(K - S_T, 0) = (K - S_T)^+.$$

The option buyer has a right. The option seller, or writer, has an obligation. This asymmetry is why the buyer pays an upfront premium.

If a call premium is c_0, the buyer's profit at maturity is

$$(S_T - K)^+ - c_0 e^{rT}$$

when we account for the opportunity cost of the premium. If we ignore financing over a short horizon, the common profit approximation is $(S_T - K)^+ - c_0$.

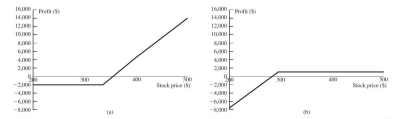

Figure 5: Option profits differ from payoffs because the option premium must be paid upfront.

European and American Exercise

A European option can be exercised only at maturity. An American option can be exercised at any time up to maturity. The labels do not refer to geography.

American exercise rights are valuable because they give the holder more choices. For a given underlying, strike, and maturity, an American option is worth at least as much as the corresponding European option:

$$C_{\text{American}} \geq C_{\text{European}}, \qquad P_{\text{American}} \geq P_{\text{European}}.$$

The inequality can be strict when early exercise has value.

Hedgers, Speculators, and Arbitrageurs

Hedgers

A hedger uses derivatives to reduce an existing exposure. The exposure exists before the derivative trade. For example, an airline worried about jet fuel prices may use energy derivatives. An exporter worried about foreign currency receipts may use FX forwards. A fund manager worried about market declines may use index futures or put options.

The key question is not whether the derivative makes money. The key question is whether the combined position has the desired risk profile. A hedge can lose money and still be successful if the underlying exposure gains enough to offset it.

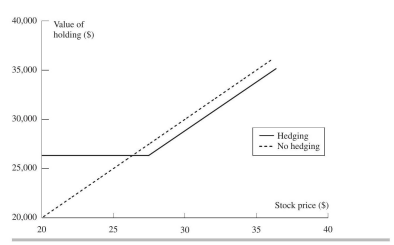

Figure 6: A hedge reduces downside exposure but may also give up upside.

Speculators

A speculator uses derivatives to take a view on the direction or volatility of a market variable. Derivatives make speculation capital-efficient because the initial cash outlay can be small relative to the notional exposure.

This is also the danger. Leverage magnifies both gains and losses. A futures position may require only margin rather than full payment for the underlying, but the economic exposure is to the full contract size.

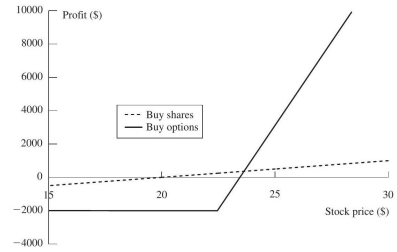

Figure 7: Leveraged derivative positions can create very different profit profiles from direct investment in the underlying asset.

Arbitrageurs

An arbitrageur tries to lock in a riskless profit by taking offsetting positions. In theory, arbitrage should require no net investment, have no risk of loss, and have a positive probability of profit. In practice, many trades called "arbitrage" are convergence trades: they are designed to profit when prices move back toward a theoretical relationship, but they can lose money before that happens.

> **Terminology warning.** True arbitrage is nearly riskless and self-financing. Convergence trading is risky. It can fail because of funding constraints, margin calls, crowded positions, model error, or liquidity shocks.

Risk Lessons from Derivatives Mishaps

Chapter 37 is useful at the start of the course because it prevents a common mistake: treating derivatives as only equations. The equations matter, but the institutions and controls matter too.

A Taxonomy of Failure

Large derivatives losses usually involve several failures at once:

1. **Market risk.** The underlying market moves against the position.
2. **Leverage.** The position is too large relative to capital.
3. **Liquidity risk.** The trader cannot exit or fund the position when needed.
4. **Model risk.** Prices, hedge ratios, or risk measures are wrong.
5. **Operational risk.** Confirmations, settlement, booking, or reporting fail.
6. **Governance failure.** Risk limits exist but are ignored, misunderstood, or overridden.
7. **Incentive failure.** Traders or managers benefit from upside while losses are borne by the institution, clients, or taxpayers.

Risk Limits

A risk limit is useful only if it is specific, measured, monitored, and enforced. Examples include notional limits, value-at-risk limits, stress-loss limits, Greek limits, stop-loss limits, maturity limits, counterparty limits, and liquidity limits.

> **Good risk limit design.** A limit should answer four questions: What exposure is being limited? How is it measured? Who sees the breach? What happens when the breach occurs?

Front, Middle, and Back Office

Financial institutions separate trading activity into three functions.

Front office Initiates trades and manages client relationships or trading books.

Middle office Measures risk, validates prices, monitors limits, and challenges assumptions.

Back office Confirms trades, settles cash flows, maintains records, and reconciles positions.

The separation is not bureaucracy for its own sake. A trader who can both take risk and control the reporting of that risk has too much unchecked authority.

Model Risk

A model is a simplified map of reality. In this course we will build models that are extremely useful: no-arbitrage pricing, binomial trees, Black–Scholes– Merton, Greeks, and volatility models. But the usefulness of a model does not make it true.

Model risk appears when:

- the wrong model is used for the product;
- parameters are estimated poorly;
- liquidity or transaction costs are ignored;
- correlations change in stress periods;
- the model is used outside the range where it was tested;
- users do not understand what the model assumes.

Rule for the course. We will derive formulas carefully, but every formula must be accompanied by its assumptions. A pricing formula without its assumptions is not knowledge; it is a recipe for misuse.

Lecture 1 Mathematical Toolkit

Positive Part Notation

For any real number x,

$$x^+ = \max(x, 0).$$

Thus the call payoff is $(S_T - K)^+$ and the put payoff is $(K - S_T)^+$.

Linear Versus Convex Payoffs

A forward payoff is linear in S_T:

$$g(S_T) = S_T - K.$$

Its slope with respect to S_T is one everywhere:

$$\frac{dg}{dS_T} = 1.$$

A call payoff is convex:

$$g(S_T) = (S_T - K)^+.$$

Ignoring the kink at $S_T = K$, its slope is

$$\frac{dg}{dS_T} = \begin{cases} 0, & S_T < K, \\ 1, & S_T > K. \end{cases}$$

This kink is the source of much of option-pricing theory. Linear exposures can often be hedged with fixed quantities. Nonlinear exposures require dynamic hedging or option-pricing models.

Simple Arbitrage Logic

Most no-arbitrage arguments follow the same structure:

1. Identify two strategies with the same future payoff.
2. Compute their current costs.
3. If the costs differ, buy the cheaper strategy and sell the expensive strategy.

4. The price discrepancy cannot persist in a competitive market.

In the forward-price derivation, the two strategies are:

buy stock now and finance it versus enter forward to buy later.

Both deliver ownership of one share at time T. Therefore their costs must be consistent.

Worked Problems

Example 3 (Forward hedge payoff)
A firm will receive EUR 10 million in three months. It enters a short forward contract at USD 1.0800 per EUR. At maturity, the spot exchange rate is USD 1.0350 per EUR.

The short forward payoff per euro is

$$K - S_T = 1.0800 - 1.0350 = 0.0450.$$

For EUR 10 million, the derivative payoff is

$$0.0450 \times 10{,}000{,}000 = \$450{,}000.$$

The receivable itself is worth fewer dollars when the euro falls. The forward gain offsets that loss. The firm has locked in

$$1.0800 \times 10{,}000{,}000 = \$10.8 \textit{ million}.$$

Example 4 (No-arbitrage forward price)
A non-dividend-paying stock has $S_0 = 50$. The continuously compounded risk-free rate is $r = 4\%$ and $T = 0.5$ years. The no-arbitrage forward price is

$$F_0 = 50e^{0.04\times 0.5} = 50e^{0.02} \approx 51.01.$$

If the six-month forward price were 53, the cash-and-carry arbitrage would be: borrow 50, buy the stock, and short the forward. The locked-in terminal profit would be

$$53 - 50e^{0.02} \approx 1.99.$$

Example 5 (Margin account)
A trader is long one futures contract on 1,000 units. The settlement price falls from 82.40 to 81.75. The daily futures P&L is

$$1 \times 1{,}000(81.75 - 82.40) = -650.$$

The long loses 650 and the short gains 650. If the long's margin account was 2,200 before settlement, it becomes 1,550 after settlement. If maintenance margin is 1,600, a margin call is triggered.

Checklist for Students

After this lecture you should be able to:

- define derivative, underlying, payoff, profit, value, notional, and market value;
- distinguish exchange-traded and OTC derivatives;
- explain the role of clearing houses, CCPs, margin, and collateral;
- write long and short forward payoffs;
- derive $F_0 = S_0 e^{rT}$ for a non-dividend-paying asset;
- explain why futures prices converge to spot prices near delivery;
- compute daily futures gains and losses from settlement prices;
- write call and put payoffs using positive-part notation;
- distinguish hedging, speculation, true arbitrage, and convergence trading;
- identify common causes of derivatives losses.

Practice Questions

1. A long forward has delivery price $K = 120$. Draw and label the payoff diagram. What is the payoff if $S_T = 95$, 120, and 138?
2. A stock has price $S_0 = 80$, pays no dividends, and the continuously compounded risk-free rate is 3%. Compute the one-year no-arbitrage forward price. Describe the arbitrage if the quoted forward price is 85.
3. A trader is short three futures contracts. Each contract is for 5,000 units. The settlement price changes from 4.62 to 4.55. Compute the daily P&L.
4. Explain why margin is not the same as the price of a futures contract.
5. A European call has strike $K = 50$ and premium $c_0 = 4$. Compute the payoff and approximate profit at maturity for $S_T = 42$, 50, and 63, ignoring financing of the premium.
6. Give one example of a hedge, one example of speculation, and one example of a trade often called arbitrage that is actually convergence trading.
7. Pick one derivatives mishap discussed in class. Identify at least three distinct risk-management failures involved.